Forest Policy and Economics 137 (2022) 102691

Contents lists available at ScienceDirect

Forest Policy and Economics

journal homepage: www.elsevier.com/locate/forpol

Perfect assumptions in an imperfect world: Managing timberland in an oligopoly market

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ARTICLE INFO

Keywords:
Regulated forest
Forest management
Game theory
Market competition

ABSTRACT

We built a game-theoretic supply model where forest landowners respond to each other's decisions using two market assumptions: (i) Perfect cartel, (ii) Cournot competition (simultaneous moves) and (iii) Stackelberg competition (sequential moves). Our findings indicate that the initial forest structure is instrumental in determining forest composition outcomes among suppliers. The solutions in the Cournot model, landowners with the same initial forest structure have uniform outcomes with increased variation in financial performance arising with different initial endowments of pulpwood and sawtimber and establishment costs. Alternatively, Stackelberg leadership has profound financial benefits to leaders even under similar initial conditions, that remain regardless of scenario. However, while terminal overall forest composition was similar regardless of scenario under Cournot outcomes, the same is not true under Stackelberg. We find that Stackelberg outcomes led to the follower being unable to harvest younger age classes over time, which resulted in accumulation of older age class stands. Our results elucidate the importance of diversification and policies that reduce landownership land concentration.

1. Introduction

Forest economics analyses and forest management planning often implicitly assume perfect atomistic competition with many buyers and sellers in a free market. The market-clearing price is achieved through multiple volunteered transactions where a single or few producers or consumers are incapable of exercising any market power. Although this assumption is realistic for specific market locations and scales, its merits have not been examined much in the forestry sector. A few studies have investigated market structure and power (oligopsony) by timber buyers, but almost none have discussed the converse situation of market power by timber growers (oligopoly). This major gap in our understanding and empirical evidence about the functioning and outcomes of timber markets under different degrees of competition can affect both timberland owners and forest products firms and their strategic and operational decisions.

While empirical analyses indicate strong evidence of different degrees of market power among the wood-consuming mills (Mead, 1966; Mei and Sun, 2008; Murray, 1995; Silva et al., 2019), on the supply side, market competition commonly classifies timber producers as price-takers (e.g., Polyakov et al., 2010). This assumption ignores that wood consuming-mills could have few timber suppliers with enough market power to alter quantities and increase negotiated prices. Among the factors that could lead to this condition are: (i) seasonal timber harvest restrictions that limit the harvesting access to specific regions, reducing the size of timber procurement area (Conrad IV et al., 2017), (ii) specialized sawmills that demand specifics species and log dimensions not easily accessible (Espinoza et al., 2011), (iii) a limited number of landowners that could provide a steady supply of timber through a long-term contract (Brodrechtova, 2015; Yin and Izlar, 2004), and, (iv) markets with a large share of public forestland with large areas and timber stock (Adams and Latta, 2005).

An oligopolistic timber market may better approximate conditions where large percentages of the forest are owned and managed by public or private entities. For instance, large shares of public land should have a greater effect on timber prices and quantity supplied. In the past in the U.S. private sector (or now in many non-U.S. countries), large vertically integrated forest products companies owned the majority of timberland,

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https://doi.org/10.1016/j.forpol.2022.102691
Received 9 April 2021; Received in revised form 31 December 2021; Accepted 5 January 2022
Available online 8 February 2022

and a small share of timber was from small private landowners. In some locations in the U.S., major Timber Investment Management Organizations (TIMOs) or Real Estate Investment Trusts (REITs) manage or own large blocks of timberland that could provide a substantial timber supply market power. These potentially major shortcomings in the assumptions of perfect markets and competition and their effects on timber growers as producers in the timber factor market, as well as wood-consuming forest products mills, are an important but little investigated subject.

Although forest planners and managers often face the preceding situations, current Decision Support Systems (DSS) for forest management have not yet explored how the competition between a few landowners could affect forest management financial returns, and strategic and tactical planning. Several questions arise when oligopoly conditions are observed, such as how silvicultural decisions change compared to price-taker assumption and how the forest structure is affected. The profits of large versus small forest landowners and of quantity leaders versus followers also may differ considerably under oligopoly timber supply conditions.

In a market with few landowners, harvest schedule decisions more closely follow game-theoretic behavior; landowners are expected to respond to each other's decisions, thereby affecting timber quantities and prices differently. These types of models have been used in many different situations, like the supply chain in agriculture (e.g., Bai et al., 2012), water resources (e.g., Madani, 2010), forest products (e.g., Yue and You, 2017), and the energy sector (e.g., Safarzadeh and Rasti-Barzoki, 2019). They assume that decision-makers optimize their outcome unilaterally or cooperatively, resulting in different results from those predicted by perfect competition. For instance, Yue and You (2017) investigated the competition between biorefineries and pulp mills for wood fiber; they reported that the profit share between biorefineries and pulp mills was the same across different game-theoretic models, while in a centralized approach was 18% higher for the pulp mills. Social welfare as measured by economic models of producer and consumer surplus can also vary between model assumptions (Brannlund, 1989; Murray, 1995). Bai et al. (2012) showed a welfare difference of more than a hundred million dollars (USD) between cooperative and non-cooperative games in a biofuel supply chain model.

Similar outcomes are expected in forest planning decisions. Sustainable forest management is based on financial, biological, and social-economic equilibrium; therefore, having different assumptions about market interactions can considerably impact forest product prices, removals, and ecological structure. Understanding this dynamic helps landowners to optimize their forest production, wood consumers to gauge their appropriate responses and strategies in wood procurement and market development, and policymakers to understand and assess forest sector market conditions. To model and examine the effect of different timber production supply structures under conditions of less than the perfect atomistic competition, we built three models of a simplified economic structure composed of just two timber producers — the opposite of a perfectly competitive production market. This helped provide much more insight into how timber production may respond to imperfect competition. Analyzing this opposite market extreme of two producers may oversimply timber markets somewhat but might well be closer to reality than atomistic competition assumptions, particularly in small wood basins.

Three models were tested using nine scenarios: (i) a perfect cartel model, (ii) a Cournot cooperative producer model, and (iii) a Stackelberg sequential market leader/follower model. We simulated a forest basin with two landowners managing a forest for 50 years. The results indicate a substantial change in the forest structure and management style according to initial forest structure and model assumptions.

2. Literature review

Forest management methods have noticeably improved over the last several decades, from the stand-level optimization to mathematical programming techniques for large-scale problems. A large share of the literature on forest management has focused on understanding the tradeoff between different forest outputs such as timber production, ecosystem services, wildfire, and pest outbreaks (e.g.,Ager et al., 2019; Eyvindson et al., 2018; Lauer et al., 2017a; Marto et al., 2018; Pohjanmies et al., 2017). The modeling community has also developed powerful DSS that include stochastic models in response to biological and market risks like wildfire (e.g., Garcia-Gonzalo et al., 2014; Lauer et al., 2017), interest rates (Zhou and Buongiorno, 2011), and stumpage prices (e.g., Brazee and Bulte, 2000; Reeves and Haight, 2000). In addition, there are efforts to integrate tactical and operational planning to optimize the supply chain and reduce possible disconnection between the forest planning and field operations (e.g., Belavenutti et al., 2019; Eyvindson et al., 2018).

Although these models are widely used in the forest sector, they often ignore local social-economic factors. To overcome this deficiency, agent-based models have included the ecology, social-economic indicators, and landowner's behavior into management decisions (e.g., Bone and Dragićević, 2010). This method has been used to examine forest management outcomes for several topics like fire management (e.g., Spies et al., 2017), bioenergy sustainability assessment (e.g., Zupko and Rouleau, 2019), timber markets (Henderson and Abt, 2016; Sjolie et al., 2011), land-use change (Bakker et al., 2015) and ecosystem services (Sotirov et al., 2019)).

These DSS have covered a large spectrum of economics and mathematical optimization techniques; however, there are few examples where forest managers could simulate the interactions among market players, like the ones predicted by game theory. Forest product models that explicitly use game theory frameworks focus commonly on auctions (e.g., timber - Athey and Levin, 2001; Baldwin et al., 1997; Haile, 2001 - or ecosystem services - Reeson et al., 2011). Paradis et al. (2018) and Paradis et al. (2013) investigated the disconnection of strategic and operational forest planning and simulated them simultaneously using a bilevel hierarchical model. These studies emphasized the impact of hierarchical decisions on the sustainability of forest business in the long run. Yue and You (2017) developed a hierarchical model to identify an optimal biomass plant location based on the interaction with pulp mills and sawmills and landowner returns. Similar to Paradis et al. (2018) and Paradis et al. (2013), Yue and You (2017) did not study competition between forest landowners; instead, they investigated the role of a biofuel mill as a fiber consumer "leader" followed by pulp mills and other market players.

Our study extends the existing literature in several ways: First, we developed a model framework for simulating competition among forest landowners in stumpage markets under less than perfect competition. Second, we demonstrated how a different market structure than the perfect competition could change the optimal harvest levels and inventories, as well as profits. Third, while we focused on examining different market structures, our solution approach is the first to be applied with and incorporated into harvest schedule models. In fact, instead of an aggregate econometric or biological model of timber markets under perfect competition, we examine market structure effects on forest management decisions. Last, this set of theoretical, empirical, and computational advances adds a new set of tools for the academic and private community for research study and planning of forest products markets and ecological structure.

3. Theoretical approach

As noted, our models provide new means to analyze the effect of different timber market conditions other than perfect competition. Our central assumption is that timber suppliers will select the quantities and silvicultural treatments that maximize profit and that timber prices will be derived from the perfect cartel, Cournot, or Stackelberg quantity leadership models. To facilitate the interpretation of the theory and linkage to our numerical example, we focus only on two producers (a duopoly), but the problem can be expanded to include multiple landowners given the optimality conditions are met. Before delving into Cournot and Stackelberg, we explore the extreme situation of complete cooperation between competitors as in a perfect cartel.

3.1. Perfect cartel model

In the perfect cartel, landowners one and two choose the quantity (q_1 and q_2) that maximizes their profits in complete cooperation such as in:

$$\boldsymbol{\pi}_1 + \boldsymbol{\pi}_2 = q_1 \times P(Q) - \boldsymbol{C}_1(q_1) + q_2 \times P(Q) - \boldsymbol{C}_2(q_2) \tag{PC1}$$

where $\boldsymbol{P}(Q)$ is the inverse demand function, Q is the total market supply $(q_1 + q_2)$, $\boldsymbol{C}_{1(2)}(q_{1(2)})$ is a cost function of each landowner. In the optimum, the first-order conditions of Eq PC1 in respect to q_1 and q_2 are defined as:

$$\underbrace{\overbrace{q_1 \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right)}^{A_1} + \boldsymbol{P}(Q)}_{MR_1} - \underbrace{\frac{\partial \boldsymbol{C}_1(q_1)}{\partial q_1}}_{MC_1} = \underbrace{\overbrace{q_2 \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right)}^{A_2} + \boldsymbol{P}(Q)}_{MR_2} - \underbrace{\frac{\partial \boldsymbol{C}_2(q_2)}{\partial q_2}}_{MC_2} \tag{PC2}$$

where MR_1 and MR_2 are the marginal revenues, and MC_1 and MC_2 are the marginal costs. Notice that the total marginal revenue in this model is higher than in a perfectly competitive market $\left(\boldsymbol{P}(Q) - \frac{\partial C(q_{1(2)})}{\partial q_{1}(2)} = 0\right)$ because of the terms $A_{1(2)}$. An additional unity on top of $q_{1(2)}$ causes a decrease in prices by the term $A_{1(2)}$ since the demand function has a downward slope. Adjusting Eq. PC2, we have the response function of firm one (or two) represented as:

$$\boldsymbol{f}_{1(2)}(.) = \boldsymbol{q}_{1(2)} = \left(q_{2(1)} \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right) - \frac{\partial \boldsymbol{C}_{2(1)}\left(q_{2(1)}\right)}{\partial q_{2(1)}} + \frac{\partial \boldsymbol{C}_{1(2)}\left(q_{1(2)}\right)}{\partial q_{1(2)}}\right) \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right)^{-1} \tag{PC3}$$

Notice that the quantity provided by firm one (q_1) is a response to quantity provided by firm two (q_2) defined in the market equilibrium (Q) as well as their marginal costs.

3.2. Cournot model (simultaneous market decisions)

The Cournot model assumes that the quantity produced by a firm one (q_1) is a simultaneous response to the quantity offered by its competitors (q_2). Considering a market where firm one is a profit maximizer such that:

$$\boldsymbol{\pi}_1 = q_1 \times P(Q) - \boldsymbol{C}_1(q_1) \tag{C1}$$

The first-order condition of $\boldsymbol{\pi}_1$ in respect to q_1 is defined as:

$$\underbrace{\overbrace{q_1 \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right)}^{A} + \boldsymbol{P}(Q)}_{MR} - \underbrace{\frac{\partial \boldsymbol{C}_1(q_1)}{\partial q_1}}_{MC} = 0 \tag{C2}$$

Now, the marginal revenue (MR) is still higher than a perfectly competitive market, but only by the term A, not by the combination of both landowners as in the perfect cartel. Thus, the effect on prices is less than in the perfect cartel. Similarly, to the perfect cartel, the response function of firm one (or two) is represented as:

$$\boldsymbol{f}_{1(2)}(.) = \boldsymbol{q}_{1(2)} = \left(\frac{\partial \boldsymbol{C}_1\left(q_{1(2)}\right)}{\partial q_{1(2)}} - \boldsymbol{P}(Q)\right) \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q}\right)^{-1} \tag{C3}$$

There is an important change in the Cournot model response function (Eq. C3) in comparison to the perfect cartel (Eq. PC3). Instead of being affected by the marginal cost of the competitors and the total quantity supplied, the response function of each landowner is now impacted by the market price defined in equilibrium and only their own marginal costs.

3.3. Stackelberg model (quantity leadership)

In the Cournot model, firms are assumed to behave as if the supply decision is taken simultaneously; however, there are cases that firms might know beforehand about the competitors' supply quantity and make their decision as a market leader. In the Stackelberg model, firms are divided into leaders and followers, where leaders optimize their profit based on the follower's optimal decisions. If firm one behaves as a leader, its profit maximization is defined as:

$$\boldsymbol{\pi}_1 = q_1 \times \boldsymbol{P}(q_1, \boldsymbol{f}_2(.)) - \boldsymbol{C}_1(q_1) \tag{S1}$$

where $\boldsymbol{f}_2(.)$ is a response function from firm two derived from Eq. C3. Now, the quantity supplied by the leader (firm one) in response to the optimal provides by the follower (firm 2) is derived from the first-order condition as in:

$$\boldsymbol{q}_{1(2)} = \left(\frac{\partial \boldsymbol{C}_{1(2)}\left(q_{1(2)}\right)}{\partial q_{1(2)}} - P(Q)\right) \times \left(\frac{\partial \boldsymbol{P}(Q)}{\partial Q} + \frac{\partial \boldsymbol{P}(Q)}{\partial Q}\left(1 + \frac{\partial \boldsymbol{f}_{2(1)}(.)}{\partial q_{1(2)}}\right)\right)^{-1} \tag{S2}$$

The interpretation of Eq. S2 is similar to the Cournot model (C3); however, they have an additional marginal effect from the follower's response function ($\frac{\partial f_{2(1)}(.)}{\partial q_{1(2)}}$). This marginal response function has two primary consequences: (i) the quantity produced per firm is not necessarily the same even under equal marginal costs, and (ii) prices are even further from the perfect competition assumption given the additional marginal response function. We expect that profits in the Stackelberg model will be higher for the leader than in the Cournot setup since leaders could choose their production based on follower's decisions and not an arbitrary quantity as in the Cournot. Choosing between being a leader or a follower will depend on some conditions: *(i)* wood-mills have to be indifferent about different suppliers, and *(ii)* the supplier response functions are downward sloping, in other words, landowner 1 (2) will reduce/increase the quantity supplied if its competitors increase/reduce their quantity supplied.

4. Integrated timber market structure and forest stand model

To test the impact on forest landowner outcomes within the aforementioned market structures, we use key forest data sources to develop a detailed representation of the landowner strategies within simulated theoretical scenarios.

4.1. Variable description and notation

In our model, forest landowners one and two ($j \in \{1, 2\}$) maximizes profits based on a set of silvicultural treatments b over the following T periods. Landowners perfectly foresee the demand. The quantity of timber supplied and inventory of product k by landowner j at period t are denoted respectively as $q^s_{jkt} = \sum_b^B l_{kbit} X_{jbvt}$ and $\boldsymbol{g}_{jkt} = \sum_b^B l_{bt} Z_{jbvt}$, where

l_{kbit} is the volume per acre of product k under silvicultural treatment b at age i and period t, and X_{jbvt} and Z_{jbvt} are, respectively, the area harvested and inventory of landowner j allocated to treatment b, stand v at period t. The costs by age i are denoted as c_i and the total costs per landowner j at period t is derived from $C_{jt} = \sum_v^V \sum_b^B \sum_i^I c_i Z_{jibvt}$. Notice that costs are linked only to the area managed; they could also be related to inventory or harvesting volume; however, we assume that timber is negotiated at stumpage prices, and loggers or contractors would provide harvesting operations and other costs at a fixed rate. In addition, we did not include the spatial location of the mills and landowners, which would directly affect the management intensity, and thus the inventory composition. The spatial component is the next step to this type of model.

We denoted the wood inventory g_{ikt} as the upper bound limit of the timber production q^s_{jkt}. The market supply is thus the sum of all landowners' harvest volumes, $\sum_{j=1}^J q^s_{jkt}$, and constrained by the total market equilibrium conditions $Q^s_{kt} = Q^d_{kt}$, where Q^s_{kt} and Q^d_{kt} are, respectively, the market supply and demand of product k at period t. The price of the product k during period t is defined by the inverse demand function $p_{kt} = \alpha_k (Q^d_{kt})^{\beta_k}$ where $\beta_k \leq 0$ is the price elasticity of demand by product.

4.2. Perfect cartel

The perfect cartel model assumes that landowners maximize profits in cooperation. We represent this model mathematically as:

$$Maximize \sum_j^J \sum_k^K \sum_t^T \left(p_{kt} q^s_{jkt} - C_{jt} \right) / (1+\delta)^t \tag{1}$$

subjected to:

$$q^s_{jkt} = \sum_v^V \sum_b^B \sum_i^I l_{kbit} X_{jibvt} \ \forall j \in \{1,2\},\ k \in K,\ t \in T \tag{2}$$

$$g_{jkt} = \sum_v^V \sum_b^B \sum_i^I l_{kbit} Z_{jibvt} \ \forall j \in \{1,2\},\ k \in K,\ t \in T \tag{3}$$

$$X_{jibvt} \leq Z_{jibvt} \ \forall j \in \{1,2\},\ i \in I,\ b \in B,\ v \in V,\ t \in T \tag{4}$$

$$C_{jt} = \sum_v^V \sum_b^B \sum_i^I c_i Z_{jibvt} \ \forall j \in \{1,2\},\ t \in T \tag{5}$$

$$q^s_{jkt} \leq g_{jkt} \ \forall j \in \{1,2\},\ k \in K,\ t \in T \tag{6}$$

$$Q^s_{kt} = \sum_j^J q^s_{jkt} \ \ \forall k \in K,\ t \in T \tag{7}$$

$$Q^s_{kt} = Q^d_{kt} \ \forall k \in K,\ t \in T \tag{8}$$

$$p_{kt} = \alpha_k \left(Q^{s=d}_{kt}\right)^{\beta_k} \ \forall k \in K,\ t \in T \tag{9}$$

$$q^s_{jkt} \geq 0 \ \forall j \in \{1,2\}, k \in k, t \in T \tag{10}$$

$$p_{kt} \geq 0 \ \forall k \in K,\ t \in T \tag{11}$$

$$X_{jibvt} \geq 0;\ Z_{jibvt} \geq 0 \ \forall j \in \{12\},\ i \in I,\ b \in B,\ v \in V,\ t \in T \tag{12}$$

where Eq. 1 represents the total net present value maximization at a discount rate δ, Eqs. 2 and 3 are, respectively, the volume of timber harvested and inventory of landowner j, product k at period t. Eq. 4 guarantees the area harvested will not be higher than the area available. Eq. 5 is the total cost per producer and period. Eqs. 6 restricts the quantity harvested to less or equal to the inventory available. Eqs. 7, 8, and 9 are respectively the timber supply, the equilibrium condition, and the inverse demand function. Eqs. 10 to 12 guarantee their respective variable are positive.

4.3. Cournot model

In the Cournot market model, landowner 1 (or 2) defines its production $\left(q^s_{(j=1(2))kt}\right)$ based on profit maximization under a discount rate (δ) simultaneously to its competitor outputs ($f^s_{(j=2(1))kt} = q^s_{(j=2(1))kt}$), mathematically:

$$Maximize \sum_k^K \sum_t^T \left(p_{kt} q^s_{jkt} - C_{jt} \right) / (1+\delta)^t \ \forall j \in \{1,2\} \tag{13}$$

subjected to:
Eqs. (2) to (12)

To guarantee that a landowner 1 is maximizing profit based on the best response function to their competitors ($q^s_{(j=1(2))kt}$), like proposed in theory, we imposed the Karush-Kuhn-Tucker (KKT) optimal conditions. Because the Stackelberg Model also requires this arrangement, we introduce the KKT concepts later in the paper.

4.4. Stackelberg model (quantity leadership)

Here, we use the Stackelberg Model where landowner 1 acts as a "leader" dominating the market based on landowner 2, "follower", decisions. That is, landowner 1 will maximize profit based on the landowner's two best response functions. We represent the Stackelberg market mathematically as:

$$Maximize \sum_t^T \sum_k^K \left(p_{kt} q^s_{(j=1)kt} - C_{(j=1)t} \right) / (1+\delta)^t \tag{14}$$

Subjected to:

$Eqs.(2)\ to\ (6)\ \ \forall\ j = 1,\ k \in K,\ t \in T$

Maximize

$$\sum_k^K \sum_t^T \left(p_{kt} q^s_{(j=2)kt} - C_{(j=2)t} \right) / (1+\delta)^t \tag{15}$$

Subjected to:
Eqs. (2) to (6) $\forall j = 2,\ i \in I,\ b \in B,\ k \in K,\ v \in V,\ t \in T$
Eqs. (7) to (12) $\forall j \in \{1,2\},\ i \in I,\ b \in B,\ k \in K,\ v \in V,\ t \in T$

Notice that the problem follows a hierarchical structure, as in a Bilevel Optimization model, and the leader defines its strategy based on the follower profit maximization (Eq. 15).

5. Estimation

We solved the harvest scheduling optimization presented above using Gurobi 9.2. Gurobi uses a McCormick Relaxation in the spatial branch and bound algorithm to remove the violations of the nonconvexities presented in the profit function (more details at Tsoukalas and Mitsos, 2014). The Cournot and Stackelberg models were reformulated following the Karush-Kuhn-Tucker (KKT) optimal conditions as follows.

5.1. Karush-Kuhn-Tucker (KKT) reformulation

We use the KKT reformulation to (1) guarantee the quantities from the response functions in the Cournot model are at their optimal, and (2) transform the bilevel optimization problem proposed in section 4.3 into a Mixed-Integer Programming (MIP) problem. The KKT reformulation is obtained by introducing constraints that ensure the optimal conditions: stationarity, primal feasibility, and complementary slackless (Hobbs, 2001; Colson et al., 2007; Sinha et al., 2018). Notice that the notation for Cournot and Stackelberg are slightly different; whereas in the Cournot formulation there are two players in the set of firms ($j \in \{1,2\}$), in the Staclekberg, the KKT conditions are applied only to the follower ($j = 2$). The reformulation of the Cournot and Stackelberg models proposed in section 4.2 and 4.3 expands Eqs. 1 and 14 (Profit Maximization), Eq. 6 (Inventory Constraint), Eq. 8 (Market Equilibrium), Eq. 9 (Price at equilibrium), and Eq. 10 (Positive supply), to Eqs KKT 1, KKT 2, KKT 3, and KKT 4 for both models separately.

$$\frac{\left[\frac{\partial p_{kt}\left(q^s_{jkt}\right)}{\partial Q^s_{kt}} \boldsymbol{q}^s_{jkt} + \boldsymbol{p}_{kt}\left(\boldsymbol{Q}^s_{kt}\right)\right]}{(1+\delta)^t} - \lambda^1_{jkt} - \mu^1_{kt} - \mu^2_{kt}\frac{\partial \boldsymbol{p}_{kt}\left(\boldsymbol{Q}^s_{jkt}\right)}{\partial \boldsymbol{Q}^s_{kt}} + \lambda^2_{jkt} = 0 \tag{KKT 1}$$

$$\lambda^1_{jkt}\left(\boldsymbol{q}^s_{jkt} - \boldsymbol{g}_{jkt}\right) = 0 \tag{KKT 2}$$

$$\lambda^2_{jkt}\boldsymbol{q}^s_{jkt} = 0 \tag{KKT 3}$$

$$\lambda^1_{jkt} \geq 0; \lambda^2_{jkt} \geq 0 \tag{KKT 4}$$

where, $\frac{\partial p_{kt}\left(q^s_{jkt}\right)}{\partial Q^s_{kt}} = \alpha_k\beta_k\left(Q^{s=d}_{kt}\right)^{(\beta_k - 1)}$, λ^1_{jkt}, $\mu^1_{kt}, \mu^2_{kt}, \lambda^2_{jkt}$ are the Lagrange multipliers of Eqs. 6,8, 9, and 10, respectively. The constraints presented in Eq KKT 1 are the stationary conditions; constraints in Eq KKT 2 ensure primal feasibility of the lower level; Eq KKT 3 are the complementary slackness, and condition Eq KKT 4 is the dual feasibility.

The reformulation above brings some limitations since Eqs KKT 2 and KKT 3 introduce a non-linear complementary constraint, which can bring computational challenges. These complementary constraints can be transformed into Mixed-Integer-Constraints (MIC) (Garcia-Herreros et al., 2016) by first introducing a slack variable (s_{jkt}) in Eq. 6 and rewrite it as.

$$\boldsymbol{q}^s_{jkt} + s_{jkt} = \boldsymbol{g}_{jkt} \tag{MIC 1}$$

Next, we use the Big-M reformulation to impose that the constraint in Eq KKT 2 or its corresponding multiplier (λ^1_{jkt}) is "activate" as follows:

$$\begin{aligned} &s_{jkt} \leq M^A_{jkt}w^1_{jkt} \\ &\lambda^1_{jkt} \leq M^A_{kt}\left(1 - w^1_{jkt}\right) \\ &w^1_{jkt} \in \{0,1\} \end{aligned} \tag{MIC 2}$$

Similarly, the Big-M reformulation of Eq KKT 3 is represented as.

$$\begin{aligned} &\boldsymbol{q}^s_{jkt} \leq M^B_{jkt}w^2_{jkt} \\ &\lambda^2_{jkt} \leq M^B_{kt}\left(1 - w^2_{jkt}\right) \\ &w^2_{jkt} \in \{0,1\} \end{aligned} \tag{MIC 3}$$

where w^1_{jkt} and w^2_{jkt} are binary variables that if equal one Lagrange multipliers λ^1_{jkt} and λ^2_{jkt} equals zero, and if equal zero, the supply variables $\boldsymbol{q}^s_{jkt}$ equals zero. $\boldsymbol{M}^{A\ (B)}_{jkt}$ is the Big-M parameter with a value equals to the maximum timber inventory. There are many forms to estimate the value $\boldsymbol{M}^{A\ (B)}_{jkt}$, some used advanced mathematical techniques (like in Appendix D in Feng et al., 2020), and others follow intuition and common sense. $\boldsymbol{M}^{A\ (B)}_{jkt}$ should be small enough that the model is still feasible, but not too large that could result in numerical issues. We assume that $\boldsymbol{M}^A_{jkt}$ in the Cournot and Stackelberg model is the highest stock volume by landowner j and product k in over 50 years found in the optimal solution of the perfect cartel model ($\boldsymbol{M}^{A\ (B)}_{jkt} = max\left[\widetilde{g}^{PC}_{(j=1)kt}, \widetilde{g}^{PC}_{(j=2)kt}\right]$). And $\boldsymbol{M}^B_{jk}$ is equal to the total demand of product k at period t in equilibrium ($Q^{s=d}_{kt}$). Lastly, to solve the duopoly problem proposed here, we replace Eq. 6, KKT 2, and KKT 3 with Eqs MIC 1, MIC 2, and MIC 3. Therefore, Cournot and Stackelberg formulations become a Mixed Integer Problem (MIP) summarized respectively as:

5.2. Cournot - MILP formulation

$$\frac{\left[\frac{\partial p_{kt}\left(q^s_{jkt}\right)}{\partial Q^s_{kt}} q^s_{jkt} + p_{kt}\left(Q^s_{kt}\right)\right]}{(1+\delta)^t} - \lambda^1_{jkt} - \mu^1_{kt} - \mu^2_{kt}\frac{\partial p_{kt}\left(Q^s_{jkt}\right)}{\partial Q^s_{kt}} + \lambda^2_{jkt} = 0\ \forall j \in \{1,2\},$$

$k \in K,\ t \in T$

Eqs.(2) to (5)

Eqs.(7) to (12)

Eq (KKT4)

Eqs (MIC 1) to (MIC 3)

5.3. Stackelberg - MILP formulation

$$Maximize\left(\sum_{t}^{T}\sum_{k}^{K}\boldsymbol{p}_{kt}\boldsymbol{q}^s_{(j=1)kt} - \mathbf{C}_{(j=1)t}\right)\Big/(1+\delta)^t$$

Subjected to:

Eqs.(2) to (5)

$$\frac{\left[\frac{\partial p_{kt}\left(Q^s_{kt}\right)}{\partial Q^s_{kt}} \boldsymbol{q}^s_{(j=2)kt} + \boldsymbol{p}_{kt}\left(\boldsymbol{Q}^s_{kt}\right)\right]}{(1+\delta)^t} - \lambda^1_{(j=2)kt} - \mu^1_{kt} - \mu^2_{kt}\frac{\partial \boldsymbol{p}_{kt}\left(\boldsymbol{Q}^s_{jkt}\right)}{\partial \boldsymbol{Q}^s_{kt}} + \lambda^2_{(j=2)kt} = 0$$

Eqs.(2) to (5)

Eqs.(7) to (12)

Eq (KKT4)

Eqs (MIC 1) to (MIC 3)

6. Data and assumption

We numerically estimated the outcomes of the beforementioned models by simulating a Pine timber (*Pinus taeda*) market with only two timber suppliers (duopoly) that maximize Net Present Value (NPV) for the next 50 years at a discount rate of 4% per year.[1] In this market, the demand for sawtimber is 607.5 thousand tons per year, while for pulpwood is 742.5 thousand million tons. We initially assumed a constant demand and analyzed how changing its value would affect the results. These demands are based on the share of both markets in the U. S. South, with on average 55% of the production flowing to pulp and processing mills and 45% to sawmills (USDA Forest Service, 2021). The initial forest inventory mimics a normal forest with 41 stands of equal area across the ages 0 to 40 years old. A normal forest is often an indicator of stable production; thus, deviation from this condition might indicate a lack of long-term sustainability. Operational costs in the first stand ages are described in (Table 1); they are similar to the forest management costs in the U.S. South. In addition, we restricted our analysis to one specie and two products to facilitate interpretation.

The silvicultural prescriptions for loblolly pine are:

Tr 0: No planting. In this treatment, the landowner chooses not to plant trees, therefore, not to make the selected area available to supply timber.

Tr 1: Clearcutting at the optimal age.

Tr 2: One thinning – reducing the basal area to 60 square feet at ages 18 and clearcutting at the optimal age.

Tr 3: One thinning – reducing the basal area to 60 square feet at ages 14 and clearcutting at the optimal age.

Tr 4: Two thinning – (1st) reduction of basal area to 60 square feet at 14 years old, (2nd) reduction of the basal area to 60 square feet at

[1] To eliminate any possible impact from the terminal values, we run the models for 100 years and presented only the results for the first 50 years.

Table 1
Operational costs in the first stand ages.

Age (years)	Cost ($/acre)
0	100
1	50
2	50
3	25

20 years old, and clearcutting at the optimal age.

Yield curves are simulated by the PTAEDA (Burkhart et al., 2008) (Fig. 1), with the initial spacing of 606 trees per acre and site index of 60 ft. The annual volume by treatment starts at age ten and is simulated until age 60.

Last, the discount rate used for each producer was 4%, and elasticities from the inverse price function ($\boldsymbol{p}_{kt} = \alpha_k\left(Q_{kt}^{s=d}\right)^{\beta_k}$) are -0.5 for pulpwood and sawtimber, and the intercept shifter α equals to 0.90 for pulpwood and 1.47 for sawtimber.

7. Scenarios

We changed three parameters to simulate nine scenarios: (i) Demand, (ii) Initial Inventory, and (iii) Discount Rate (Table 2). Demand varies 20% above or below the baseline scenario, while the proportion of sawtimber and pulpwood in the inventory conditions goes from 25% to 75%. Last, we increase the discount rate from the original 4% to 8% for both landowners and, also present a scenario where just Landowner 1 has higher discount rate. Notice we did not include the scenario which both landowners have 25% of their inventory concentrated in pulpwood, and 75% in sawtimber, because this initial structure is not feasible under the demand assumption proposed.

In the Stackelberg leadership, the leader in a market is typically linked to its size and scale; however, our approach is based on information quality and availability. Here, the leader is always better informed than the followers, which between two large landowners may represent a better research team or closer relation to wood-consuming mills. We recognize that it is not likely a landowner will be a leader for 50 years; a dynamic model that simulates the change among perfect cartel, Cournot and Stackelberg leaderships would capture these movements. However, since the movement among game assumption seems a little arbitrary due to the years of leadership, the scope of this paper focuses on a single optimization and partially address these dynamics by adding the option of not planting (Treatment 0), which reveals when players should leave the market and/or reduce their size.

8. Results

We built three harvest schedule models under a non-competitive market that assumes perfect cartel, Cournot, and the Stackelberg quantity leadership. This section shows the main results and discusses

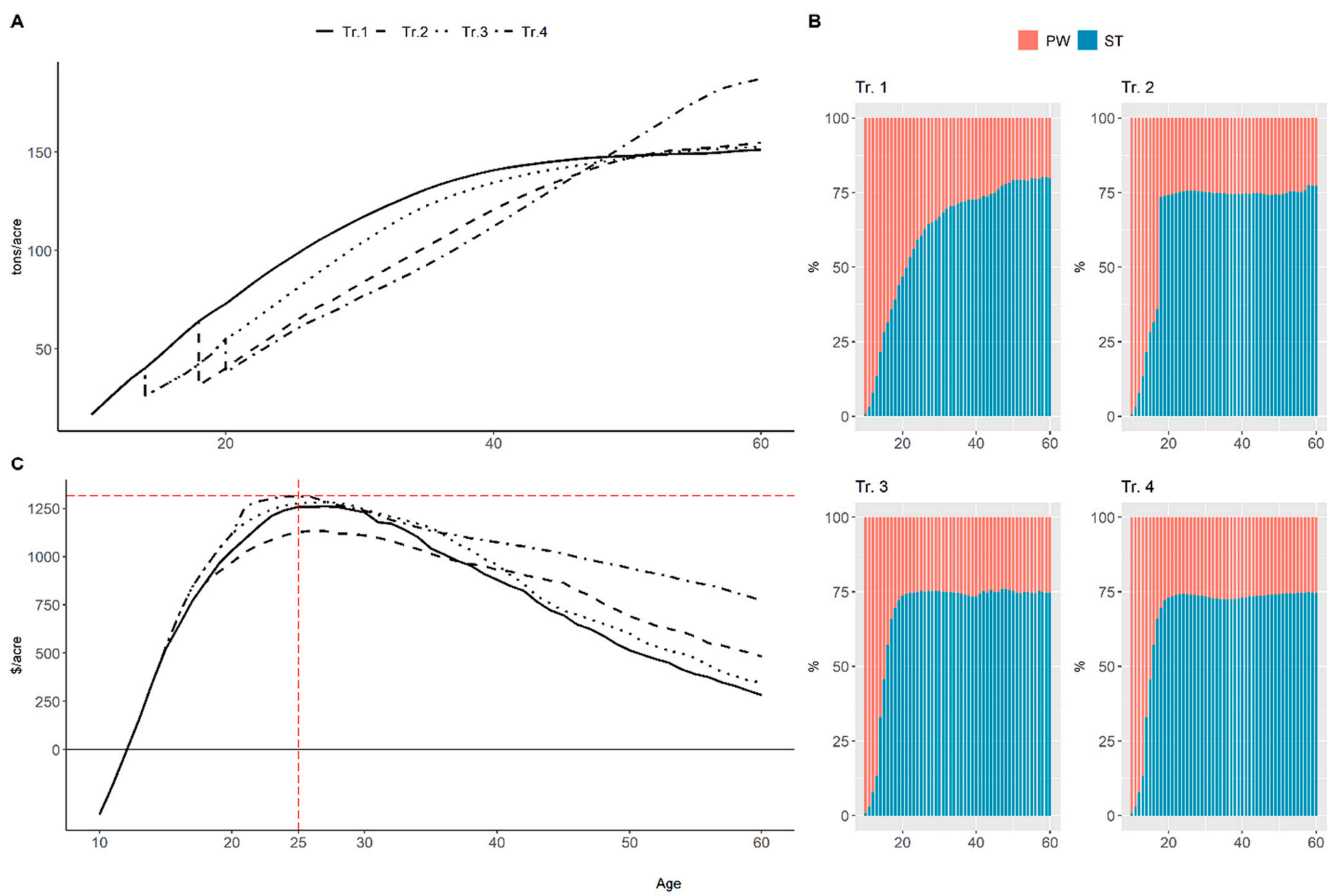

Fig. 1. Yield Curves for treatment 1 to 4. Source: PTAEDA (Burkhart et al., 2008), the initial spacing of 606 trees per acre and site index of 60 ft. Tr 1 = Clearcutting at the optimal age, Tr 2 = One thinning – reducing the basal area to 60 square feet at ages 18, Tr 3 = One thinning – reducing the basal area to 60 square feet at ages 14. Tr 4 = Two thinning – reduction of basal area 60 square feet at 14 years old, a second thinning following the same intensity at 20 years old. **(A)** Volume per acre and age, **(B)** Share of the total volume classified as sawtimber (ST) and pulpwood (PW) by age and treatment, **(C)** Land Expectation Value (LEV) at 4% discount rate by treatment and age; the intersection between the red dashed lines is the point of maximum LEV. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2
Scenarios for the demand, initial inventory conditions and discount rates.

Scenarios	Demand (thousand ton per year)		Initial Inventory	Discount Rate		
	PW	ST	Landowner 1	Landowner 2	Landowner 1	Landowner 2
S0	742.5	607.5	Normal	Normal	4%	4%
S1	742.5	607.5	Normal	25% Pulpwood 75% Sawtimber	4%	4%
S2	742.5	607.5	Normal	75% Pulpwood 25% Sawtimber	4%	4%
S3	742.5	607.5	25% Pulpwood 75% Sawtimber	75% Pulpwood 25% Sawtimber	4%	4%
S4	742.5	607.5	75% Pulpwood 25% Sawtimber	75% Pulpwood 25% Sawtimber	4%	4%
S5	20% increase	20% increase	Normal	Normal	4%	4%
S6	20% decrease	20% decrease	Normal	Normal	4%	4%
S7	742.5	607.5	Normal	Normal	8%	8%
S8	742.5	607.5	Normal	Normal	8%	4%

Note: The method used to achieve the volume proportions in the initial inventory and their initial age distribution is described in Appendix A and C respectively.

critical points such as harvesting dynamics and financial returns. We compare the results between landowners within the proposed scenarios and between the models within a scenario. Additional data on volume harvested and inventory and area by treatment are available on the journal website.

8.1. Timber harvested

8.1.1. Total Volume Harvested - comparison between the scenarios within the same model

Table 3 presents the total timber harvested in each scenario and market. The first row has the volume harvested in scenario S0 (where landowners 1 and 2 have a normal forest), while the S1 to S8 show the difference in percentage between the outcome in S0 and each scenario within the same market assumption. For example, the results in scenario S1 (where landowner 2 has 25% of their initial inventory in pulpwood and 75% in sawtimber) show that the volume harvested of pulpwood and sawtimber by landowner 1 was respectively 23.2% and 23.5% greater than in S0. On the other hand, landowner 2 harvested 23.8% and 23.7% less pulpwood and sawtimber in S1 than in S0. We observed an opposite trend in S2 (where landowner 2 has 75% of their initial inventory allocated to pulpwood and 25% to sawtimber); there was a greater supply of pulpwood and sawtimber by landowner 2 (25.6% of pulpwood and 17.9% of sawtimber) and a lower supply by landowner 1 (25% of pulpwood and 17.8% of sawtimber) than in S0. The gap to S0 is even wider in S3 (where both landowners have 75% of their inventory in pulpwood and 25% in sawtimber); landowner 2 supplied additional 34.6% of pulpwood and 33.1% of sawtimber in S3, while landowner 1 harvested 33.7% and 32.8% less pulpwood and sawtimber respectively than in S0. This harvest dynamic in which the landowner with lower initial pulpwood inventory tend to harvest less in comparison to S0 is not observed in either the Cournot or in the Stackelberg model.

In the Cournot model, scenario S1, landowner 2 has initially higher concentration in sawtimber (75%) leading landowner 1 to harvest 17.9% less sawtimber than in S0 while landowner 2 supplied an extra 18.6%. In S3, this compensation is uniformly shared; landowner 1 supplies 15.9% less pulpwood and 15.1% more sawtimber, while landowner 2 supplies 15.9% more pulpwood and 15.7 less sawtimber). In Stackelberg model, the advantage of having higher initial inventory volumes is eliminated due to the landowner 1 leadership. Among the scenarios with different forest structure (S1 to S4), only in S4, where both have higher concentration in pulpwood, landowner 1 tended to supply higher quantities of pulpwood (14.1% in comparison to S0) while landowner 2 increased sawtimber supply (229.3%).

Our findings indicate that the initial forest condition (S1 to S4) and changes in the demand (S5 and S6) have greater impact in the harvest levels than in the scenarios with the discount rate (S7 and S8). In fact, the changes in demand and discount rate had predictable impact on the volume harvested. The results in S5 and S6 show that increasing/decreasing the demand by 20% affected proportionally the increase/decrease in the supply in most of cases. In the perfect cartel scenario, landowner 1 expanded the supply in both products by more than 20% while landowner 2 increased 16% in S5. This expansion/contraction was more evenly distributed between landowner 1 and 2 in the Cournot model. In S5, landowner 1 increased the volume harvested of pulpwood and sawtimber by 21% and 21.7% respectively, while landowner 2 increased 19% in pulpwood and 18.2% in sawtimber; the outcomes in S6 is very similar but in the opposite direction. In contrast to the perfect cartel and the Cournot approach, in the Stackelberg formulation only landowner 1 (the leader) changed its production proportionally, +/−

Table 3
Difference in Quantity Harvested (%) - Reference: Scenario S0.

Scenarios	Perfect Cartel				Cournot				Stackelberg			
	Landowner 1		Landowner 2		Landowner 1		Landowner 2		Landowner 1		Landowner 2	
	PW	ST	PW	ST	PW	ST	PW	ST	PW	ST	PW	ST
S0 (thousand tons)	192	155	187	154	189	158	189	152	316	309	63	1
Difference in Quantity Harvested (%) - Reference: Scenario S0												
S1	23.2%	23.5%	−23.8%	−23.7%	5.1%	−17.9%	−5.1%	18.6%	−4.2%	−0.1%	20.9%	61.1%
S2	−25.0%	−17.8%	25.6%	17.9%	−15.8%	17.7%	15.8%	−18.4%	−7.1%	−0.3%	35.5%	124.3%
S3	−33.7%	−32.8%	34.6%	33.1%	−15.9%	15.1%	15.9%	−15.7%	−22.5%	−0.4%	113.1%	204.7%
S4	12.5%	15.4%	−12.8%	−15.5%	0.5%	1.2%	−0.5%	−1.2%	14.1%	−0.5%	−71.2%	229.3%
S5	23.9%	23.8%	16.0%	16.1%	21.0%	21.7%	19.0%	18.2%	22.3%	20.1%	8.2%	−37.9%
S6	−20.5%	−21.5%	−19.4%	−18.5%	−19.6%	−16.7%	−20.4%	−23.5%	−22.7%	−20.2%	−6.5%	60.9%
S7	−2.0%	−1.9%	2.1%	2.0%	0.0%	0.4%	0.0%	−0.4%	0.0%	−0.2%	0.1%	73.8%
S8	−76.4%	−98.7%	78.4%	99.3%	2.1%	7.3%	−2.1%	−7.5%	0.0%	−0.2%	0.1%	73.8%

Note: The difference between S0 and each scenario for every landowner and products were calculated as follows: $\left(Q_{S0}^{m} - Q_{s}^{m}\right)/Q_{S0}^{m}$, where Q_{S0}^{m} is the total quantity harvested in scenario S0 under model m. And Q_{s}^{m} is the total volume harvested for each scenario and model.

20% for both products. On the other hand, landowner 2 (the follower) harvesting was positively correlated to the increasing (decreasing) demand in S5 (S6) pulpwood market (+8.2% in S5 and − 6.5% in S6), and negatively correlated to the sawtimber market (−37.9% in S5 and + 60.9% in S6), thereby compensating the lack of supply from landowner 1.

Increasing the discount rate from 4% to 8% for both landowners (S7) did not present any substantial difference in comparison to S0 in the perfect cartel. On the other hand, the asymmetric discount rate in S8 showed that landowners with high discount rate (landowner 1) tend to harvest less than landowners with lower discount rate (landowner 2) when fully collaborating. Landowner 1 harvested respectively 76.4% and 98.7% less pulpwood and sawtimber in S8 than in S0; thus, the demand is complemented by an extra 78.4% of pulpwood and 99.3% of sawtimber supplied by landowner 2. In the Cournot model, this dynamic is not as strong since theoretically both landowners supply the same harvest volume. In the extreme cases, like in S5, the difference in percentage between the volume harvested in S3 and S0 does not exceed 25%. Last, in the Stackelberg, the landowner 1 (leader) has a small difference between S8 and S0 (less than 1%), and landowner 2 (follower) supply 73.8% more sawtimber in S8 than in S0. In other words, it is more profitable for landowner 1 to avoid an increase in harvesting than to supply timber to the market, a standard feature in non-competitive markets.

Although landowner 2 showed the largest deviation from the S0 in all Stackelberg scenarios; they started already in a lower level in comparison to the other models. In S3 the quantity of sawtimber increased more than two hundred percent, but it supplied just one thousand tons in S0. Our findings indicate that once the initial condition deviates from the normal forest, landowner 2 tended to supply more timber than in S0 as observed in S3, where pulpwood supply more than doubled (113.1%).

When investigating the impact of expansion/contraction of demand (S5 and S6) in the Stackelberg, landowner 1's supply responded proportionally. In S5, the leader dominates the market and still has enough inventory to supply additional timber; therefore, the increase in demand created a greater opportunity for extra profits and reduces the amount of timber supplied by the follower (landowner 2) in comparison to S0. Landowner 2, on the other hand, had their volume of pulpwood and sawtimber going into opposite directions; they increased the supply of pulpwood while reducing sawtimber in S5 (+8.2% and − 37.9% respectively), and, in S6, it reduced pulpwood supply by 6.5% and increased sawtimber by 60.9%.

8.1.2. *Total Volume Harvested - comparison between the models within the same scenario*

Table 4 shows the results from the comparison between models within the same scenarios. The first four columns present the results of the perfect cartel for each scenario, landowner, and product. In the remaining columns, we calculate the difference in percentage from the perfect cartel to the Cournot and Stackelberg models by scenario, landowner, and product in the same fashion as in the previous subsection. For instance, in the third column, landowner 1 supplied 1.31% less pulpwood in the Cournot model than in the perfect cartel in scenario S0.

The outcomes of the Cournot model does not differ substantially from the perfect cartel in the scenario S0, the deviation of each landowner and product is less than 2%. In the other scenarios, landowners tend to supply less than in S0 when they have a different initial condition. For instance, in S2, landowner 2 has higher pulpwood inventory in the first years, so if they were cooperating with landowner 1 (like in the perfect cartel), they would supply higher volume of pulpwood (third column – Table 4). However, since in the Cournot players act as if they are responding to the optimal supply from their competitors, its volume harvested tends to be similar. Therefore, if landowner 1 is supplying higher timber volume than landowner 2 in the perfect cartel, there will be a reduction in volume harvested in landowner 1 and increase in landowner 2 under Cournot model. For example, landowner 2 supplied 6.5% and 31.9% less pulpwood and sawtimber respectively than in the perfect cartel under scenario S2 since it supplied more timber for both products in the perfect cartel. Our findings show this trend in all scenarios, especially in S8. In the perfect cartel, when landowner 1 has a higher interest rate (S8), both landowners optimize their profits by reducing drastically the timber supply from landowner 1. In the Cournot model, landowner 1 increases the volume harvested by 326.51% and 8051% in pulpwood and sawtimber respectively to balance the timber supplied between the landowners.

In the Stackelberg model, not surprisingly, landowner 1 dominated the market and harvested substantially greater than in perfect cartel in every scenario, while, consequently, landowner 2 harvested less in both products. The largest deviation from the perfect cartel is also observed in S8. Similar to the Cournot model, in the Stackelberg model, landowner 1 increases the supply of pulpwood and sawtimber by more than 597% and 14,750%. This result reinforced that the leader could be more aggressive if they require higher interest rates to sustain their profits.

8.2. *Financial returns*

Here, we compare the NPV as: (i) in the section 8.1, between scenarios within the same model (Table 5) and (ii) in section 8.2 between models within the same scenario (Table 6).

8.2.1. *Total Net Present Value - Comparison between the scenarios within the same model*

In Table 5, the first row indicates the NPV per acre by landowner

Table 4
Difference in Quantity Harvested (%) - reference: Perfect Cartel.

Scenarios	Perfect Cartel (thousand tons)				Cournot				Stackelberg			
	Landowner 1		Landowner 2		Landowner 1		Landowner 2		Landowner 1		Landowner 2	
	PW	ST	PW	ST	PW	ST	PW	ST	PW	ST	PW	ST
S0	192.0	155.0	187.0	154.0	−1.3%	1.7%	1.3%	−1.7%	64.7%	99.0%	−66.4%	−99.6%
S1	236.0	192.0	142.0	142.0	−15.8%	−32.4%	26.2%	52.8%	28.1%	60.8%	−46.7%	−99.1%
S2	144.0	128.0	235.0	182.0	10.7%	45.5%	−6.5%	−31.9%	104.0%	141.3%	−63.8%	−99.2%
S3	127.0	104.0	252.0	205.0	25.3%	74.2%	−12.8%	−37.7%	92.7%	195.0%	−46.9%	−99.0%
S4	216.0	179.0	163.0	131.0	−11.8%	−10.8%	15.6%	14.9%	67.2%	71.6%	−88.9%	−98.3%
S5	238.0	192.0	217.0	179.0	−3.6%	−0.1%	4.0%	0.1%	62.6%	93.0%	−68.7%	−99.8%
S6	152.0	122.0	151.0	126.0	−0.1%	7.9%	0.1%	−7.7%	60.2%	102.3%	−61.0%	−99.2%
S7	188.0	152.0	191.0	157.0	0.7%	4.1%	−0.7%	−4.0%	68.1%	102.6%	−67.1%	−99.3%
S8	45.0	2.0	333.0	308.0	326.5%	8051.4%	−44.4%	−54.4%	597.1%	14,750.5%	−81.2%	−99.6%

Note: The difference between the perfect cartel for each landowner and product were calculated as follows: $\left(Q^S_{PC} - Q^S_m\right)/Q^S_{PC}$, where Q^S_{PC} is the quantity harvested under perfect cartel for each scenario S, and Q^S_m is the quantity harvested in either Cournot or Stackelberg model for each scenario S.

Table 5
Difference in the Net Present Value (%) - Reference: Scenario S0.

Scenarios	Perfect Cartel			Cournot			Stackelberg		
	Landowner 1	Landowner 2	Total	Landowner 1	Landowner 2	Total	Landowner 1	Landowner 2	Total
S0 ($/acre)	756.00	685.64	720.82	665.33	685.64	666.47	1400.07	−26.58	686.74
S1	36.48%	−45.05%	−2.30%	−11.19%	5.72%	−2.72%	0.35%	105.21%	−1.68%
S2	−7.27%	5.90%	−1.01%	20.51%	−27.33%	−3.45%	0.55%	114.50%	−1.65%
S3	−28.15%	25.47%	−2.65%	10.91%	−25.82%	−7.48%	−6.17%	−102.27%	−4.31%
S4	4.50%	−13.00%	−3.82%	−3.60%	−7.97%	−5.79%	−0.49%	29.47%	−1.07%
S5	46.62%	45.96%	46.30%	52.62%	47.85%	50.23%	45.18%	−21.30%	46.47%
S6	−42.81%	−32.26%	−37.79%	−37.62%	−41.63%	−39.63%	−36.71%	32.74%	−38.05%
S7	−0.40%	−0.10%	−0.26%	1.46%	−1.09%	0.18%	−0.23%	−40.05%	0.55%
S8	−92.61%	99.40%	-**	4.12%	−1.57%	-**	−0.23%	−40.05%	-**

Note: The difference between S0 and each scenario for every landowner and products were calculated as follows: $\left(NPV_{S0}^{m} - NPV_{s}^{m}\right)/NPV_{S0}^{m}$, where NPV_{S0}^{m} is the total Net Present Value in scenario S0 under model *m*. And NPV_{s}^{m} is the total Net Present Value for each scenario and model. ** Notice that the total NPV in S8 is not calculated because of the different discount rate used by the landowners.

Table 6
Difference in the Net Present Value (%) - Reference: Perfect Cartel.

Scenarios	Perfect Cartel ($/acre)			Cournot			Stackelberg		
	Landowner 1	Landowner 2	Total	Landowner 1	Landowner 2	Total	Landowner 1	Landowner 2	Total
S0	756.00	685.64	720.82	−11.99%	−2.63%	−7.54%	85.19%	−103.88%	−4.73%
S1	1031.78	376.77	704.27	−42.73%	87.33%	−7.94%	36.16%	−114.48%	−4.13%
S2	701.02	726.07	713.54	14.37%	−33.18%	−9.82%	100.82%	−107.85%	−5.35%
S3	543.15	860.26	701.71	35.86%	−42.43%	−12.13%	141.87%	−99.93%	−6.35%
S4	790.04	596.54	693.29	−18.81%	3.00%	−9.43%	76.34%	−105.77%	−2.01%
S5	1108.42	1000.76	1054.59	−8.39%	−1.37%	−5.06%	83.38%	−102.09%	−4.62%
S6	432.37	464.42	448.40	−4.01%	−16.10%	−10.27%	104.94%	−107.60%	−5.12%
S7	752.97	684.98	718.98	−10.35%	−3.60%	−7.13%	85.52%	−102.33%	−3.96%
S8	55.88	1367.15	-**	1139.81%	−51.93%	-**	2,399.95%	−101.17%	-**

Note: The difference between the perfect cartel for each landowner and product were calculated as follows: $\left(NPV_{PC}^{S} - NPV_{m}^{S}\right)/NPV_{PC}^{S}$, where NPV_{PC}^{S} is the total Net Present Value under perfect cartel for each scenario *S*, and Q_{m}^{S} is the total Net Present Value in either Cournot or Stackelberg model for each scenario *S*. **Notice that the total NPV in S8 is not calculated because of the different discount rate used by the landowners.

(NPV/area by landowner), and aggregated (total NPV/total area), the remaining rows compare each scenario-to-scenario S0.

In the perfect cartel, the NPV of each landowner in S0 and the scenarios with different initial inventory (S1 to S3) followed the same direction as in the results observed in the quantity harvested. For instance, in S1, landowner 2 harvested 23% less than S0 in both products, therefore the returns were also lower; however, its NPV was 45% lower than in S0, while landowner 1 had 36.4% higher NPV than in S0 instead of 23% in the harvested quantities. This finding indicates that it is not only the quantity harvested that changed, but also the period each landowner harvested. In S1, landowner 1 supplied most of the timber in the market, leading landowner 2 to hold their harvested between year one and year ten (Fig. 1D – Appendix D). In S2, S3 and S4 there was no harvesting gap over time as in S1, which reduced the losses for the landowner with lower harvested volumes. In scenarios S5 to S8, there was no unexpected results, with changes in profits directly linked to changes in demand, and the reduction of NPV due to the increase in the discount rate.

In the Cournot model, the difference between S0 and S1 to S3 is linked to the increase in the quantity supplied of sawtimber. In the extreme case, like in S2, landowner 1 increased its supply of sawtimber by 17% and its profit was 20% higher than in S0. The exception was S4, which even though landowner 1 increased 1.2% the supply of sawtimber in comparison to S0, the profit still reduced 3.5%; these results is likely due to the same initial inventory structure of landowners 1 and 2, with pulpwood having 25% of the total volume share.

Our findings in the Stackelberg model indicate that landowner 1 reduces gains in S3, S4, S6, S7 and S8. In S3, landowner 1 starts with 25% concentration in pulpwood, so even with leadership it could not increase NPV in comparison to S0, since landowner 2 has a better initial forest structure. A normal forest though is not advantageous to landowner 2 in comparison to S1, S2 and S4. Landowner 2 could increase its profit in all these scenarios by more than 100% (S1 and S2) and 29% in S4. Interestingly, in S4, landowner 1 had a slight reduction in profits (0.49%) showing that leadership with a non-favorable forest structure could harm total returns. Another interesting finding is the decrease in profit of landowner 2 when there was a increase in demand (S5); in this scenario landowner 1 took advantage and profited 45.1% more than in S0, thereby supplying almost 100% of the demand, while landowner 2 reduced its profit by 21.3%. Last, landowners 1 and 2 had equal returns in S7 and S8 even though landowner 2 had a lower discount rate in S8. The rationale behind this result is that landowner 1 did not have any more opportunity to profit beyond the conditions offered in S7, implying that the discount rate of the followers did not have great effect followers' profit.

Overall, the scenarios with different forest structure (S1 to S4) had a lower total NPV than S0 in the three models: (i) perfect cartel - from 3.8% in S4 to 1% in S2, (ii) Cournot - from 2.72% to 5.8% in the Cournot, and (iii) Stackelberg - 1.07% in S4 to 1.68% in S1. In scenarios S5 to S8, the NPV behaved as expected, with a greater/lower return in the scenario with higher/reduced demand (S5/S6). In S7 both landowners had their NPV reduced since the discount rates of both increased. This outcome indicates that a normal forest could provide landowners benefits even in scenarios where there is no any cooperation.

8.2.2. *Total Net Present Value - Comparison between the models within the same scenario*

Table 6 has the comparison between the models under the same scenario.The first three columns show the NPV per acre under the perfect cartel for each scenario, and the remaining columns compare the results in Cournot and Stackelberg. We highlight here the cooperation between the landowners provided the highest total return in every scenario than in the Cournot and Stackelberg models. In the largest difference, the total NPV reduced 12.1% in Cournot scenario S3, while in the best case, Stackelberg scenario 4, the total NPV deviated −2.01% from the perfect cartel.

In the Stackelberg model, landowner 1 increases profit in every scenario, from the lowest 36.1% in S1 to to more than 2 thousand percent in S8. On the other hand, landowner 2 losses at least 100% in every scenario. This contrast is not very clear in the Cournot, which requires a more uniform harvesting volumes. In scenario S2 and S3, for example, landowner 1 increased harvesting for both products, thus increasing profits in 14.3% and 35.8% respectively, while landowner 2 reduced its harvesting volumes, and its profit dropped 42.3% and 12.13% respectively. In these scenarios both landowners must adapt their initial inventory to reflect in a similar harvest volume.

Last, cooperation seems to be attractive only when landowners have the same discount rate; the results in S8 indicate that landowners with higher discount rates could increase their profit in more than one thousand percent if they do not cooperate, landowner 1 in the Cournot model, and by more than two thousand percent in the Stackelberg if they are market leaders under the same conditions.

9. Discussion and conclusions

This study investigated the impact of non-competitive market conditions in forest management and planning over a 50-year planning horizon. We specifically looked at harvest and profit changes in three game-theoretic models under different initial forest estates, demand conditions, and discount rates in a duopoly timber market. We compared outcomes among players in a Cournot simultaneous seller approach, Stackelberg quantity leader approach and perfect cartel model. We focus our comparison on the effects of market structure and different initial forest states assuming a two-player (landowner) game.

These three approaches of analyzing timber market structure for landowners as timber producers provide some advances in market modeling at the stand level utilizing imperfect competition rather than the standard atomistic free market assumptions. They also employ a financial optimization and forest harvest scheduling approach to examine timber market decisions.

There is a stimulating discussion in the literature around timber harvest decisions, mainly surrounded by the carbon market, biodiversity, and risk analysis. However, in many cases, free-market atomistic timber stumpage market conditions are assumed as given, and imperfect competition among forest landowners as timber market producers is not considered. This research advances this literature by exploring how market structures alter outcomes in forest management decision-making that has not been explored in previous studies to the best of our knowledge.

In line with related studies (Yue and You, 2017), our findings report substantial changes in the quantities and profits when assuming different market structure. For instance, in the perfect cartel and Cournot model, forest landowners or companies tend to collaborate, as evidenced by their similar harvest shares. Although the even shares of timber supply are not strictly tied to the normal forest structure (Uusivuori and Kuuluvainen, 2005), both players chose similar treatment options. They also have comparable rotations within those treatments indicating analogous growth patterns and creating more uniformity in timber supply. However, under quantity leadership represented in the Stackelberg game, one landowner receives much higher profits, and the combined forest estates of both combined provide lower profitability than perfect cartel and Cournot. Our findings indicate that landowners are collectively better-off in the perfect cartel and Cournot games as compared to the Stackelberg leader and follower game, not only because of higher joint profits, but also forest ownership diversity represented by landowner 2 choosing to divest of some timberland holdings under some Stackelberg scenarios (see Fig. 3D).

Concentrated market structures, market strategies, and profit outcomes can alter the provision of social benefits provided by forests, such as a stable and sustainable supply of timber, varying levels of biodiversity, and reduced fire or insect risks. Therefore, although the market leaders might profit from their market advantage, risks related to wildfire or pest outbreaks might spread over every forest stock, and less timber volumes would be available in the long-run (see Fig. 2D). Imposing these oligopoly restrictions in landscape management would reduce the timber availability from the leader, eventually opening space in the market for the followers. Though not calculated here, atomistic markets with their assumed perfect competition would have a higher total producer and consumer surplus than is the case with few suppliers. But our results here indicate the amount of the potential loss that would occur when markets are imperfect. There indeed is some middle ground between perfect competition in timber markets and a duopoly with (and without) quantity leaders, which could be analyzed in the future. Public and private stakeholders who face less competitive market conditions could incentivize a healthy forest landscape through biophysical restrictions and simultaneously creating a more competitive market.

While this analysis focused on examining timber market outcomes for forest landowners in minimally competitive factor markets, there are implications for timber buyers as well. Like many wood procurement decisions, there may be some tradeoffs between always minimizing the costs of the wood fiber in the short run (e.g., by cheap wood with easy access close to the mill) versus ensuring a stable flow of reasonably priced wood (from multiple suppliers) in each wood market. Forest buyers here might also be able to minimize their costs under an oligopoly of forest landowners by seeking a large price leader, buying more, and paying somewhat more to them but less to others, and having the cheaper wood in the short run. However, this could lead to lower total wood available in their wood basket in the long run, raising long-run wood costs. In addition, of course, the pursuit of short-run costs with one dominant landowner might have adverse biodiversity, fire, or insect and disease outcomes, and from our analysis appears to discourage replanting under certain conditions.

This work provides a new foundation for future inquiry. While this model is highly stylized, it does confirm reasonable expectations about the effects of different market structures on forest management decisions and our theoretical assumptions. The limitations might be related to the parameters, which required a particular dataset not always possible in large landscapes. Also, timber demand here was exogenous, which in calculating it endogenously would change the results profoundly. In fact, mathematical transformation applied in the methods would not be possible, and other algorithms should be implemented to solve the problem (if possible). Determining the “Big-M" values in the models also could create problems in the final solution, for instance higher “Big-M" could demand great computational power, and lower “Big-M" could make the problem infeasible; therefore, users should be careful using this approach. Another interesting point is the number of producers that these models could allocate. In the formulation presented here, the perfect cartel and Cournot could accommodate as many landowners as the computer can solve, and the Stackelberg model assumes that the leaders (if we added more than one) behave as if they are a perfect cartel while the followers as if they are in a Cournot competition. Our Stackelberg formulation does not simulate a Cournot competition between Stackelberg leaders or a perfect cartel among Stackelberg followers. Last, we have not covered at all the spatial characteristics of the timber market; distance to the mill and other transportation costs that would affect financial results among players.

Subsequent research in this subject area could be made in several topics. First, common extensions of duopoly models will examine differing marginal cost structures between suppliers (acreage differences, differing distance to a mill, tons per acre differences) as well as product quality differences (corporate/non-corporate ownership). Further, incorporating buyer-side market power into the modeling design will help to examine price differences in wood basins with one mill (one owner) to multi-mill basins (one or more owners). Second, explicitly spatial models would add more nuance in the models, in which the interactions between different levels of the supply chain could be explored (Cachon and Netessine, 2006). Third, non-competitive conditions usually surge after a sizeable external shock, like, for instance, hurricanes, therefore adding stochastic components either in prices or physical damages in the timber would provide essential tools for public and private policymakers in the strategic planning. In addition, we also could examine intermediate market structures with competition levels falling between atomistic and duopoly.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We would like to show our gratitude to the International Paper for the funding. We also are grateful to the anonymous reviewers for the insightful suggestions.

Appendix A. Goal programming

1.A Normal Forest vs Sawtimber 75% and pulpwood 25%

$$Minimize \sum_{j}^{J}\sum_{k}^{K}\sum_{t}^{T} a_{jkt}^{+} + a_{jkt}^{-}\ \forall\, t = 0 \tag{A1}$$

subjected to:

$$\boldsymbol{q}_{\boldsymbol{jkt}}^{s} = \sum_{v=0}^{V}\sum_{b=0}^{B}\sum_{i=0}^{I} l_{kibt} * X_{jibvt}\ \forall\, j \in J,\ k \in K,\ t = 0 \tag{A2}$$

$$\boldsymbol{g}_{\boldsymbol{jkt}} = \sum_{v}^{V}\sum_{b}^{B}\sum_{i}^{I} l_{ktrit} * Z_{jibvt}\ \forall\, j \in J,\ k \in K,\ t = 0 \tag{A3}$$

$$X_{jibvt} \le Z_{jibvt}\ \forall\, j \in J,\ i \in I,\ b \in B,\ v \in V,\ t = 0 \tag{A4}$$

$$Z_{jibvt} = A_{jibvt}\ \forall\, j = 1,\ i \in I,\ b \in B,\ v \in V,\ t = 0 \tag{A5}$$

$$0.75 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = g_{jkt}\ \forall\, j = 2,\ k = 2,\ t = 0 \tag{A6}$$

$$0.25 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = g_{jkt}\ \forall\, j = 2,\ k = 1,\ t = 0 \tag{A7}$$

$$\sum_{k}^{K} \boldsymbol{g}_{(\boldsymbol{j=2})\boldsymbol{kt}} = \sum_{k}^{K} \boldsymbol{g}_{(\boldsymbol{j=1})\boldsymbol{kt}}\ \forall\, t = 0 \tag{A8}$$

$$\sum_{b}^{B} Z_{jibvt} \le 400\ \forall\, j = 2,\ i \le 40,\ v \in V,\ t = 0 \tag{A9}$$

$$Z_{jibvt} = 0\ \forall\, j = 2,\ i > 40,\ b \in B,\ v \in V,\ t = 0 \tag{A10}$$

$$X_{jbt} \ge 0; Z_{jbt} \ge 0\ \forall\, j \in J,\ b \in B,\ t = 0 \tag{A11}$$

where A_{jibvt} is the initial area (Normal Forest). Eq. A9 guarantees the area younger than 40 are no larger than 40. Eq. A10 ensures there is no forest older than 40.

2A Normal Forest vs *Sawtimber 25% and Pulpwood 75%.*

Substitute Eqs. A6 and A7 by Eqs. A12 and A13 respectively.

$$0.25 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = \boldsymbol{g}_{\boldsymbol{jkt}}\ \forall\, j = 2,\ k = 2,\ t = 0 \tag{A12}$$

$$0.75 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = \boldsymbol{g}_{\boldsymbol{jkt}}\ \forall\, j = 2,\ k = 1,\ t = 0 \tag{A13}$$

3A Sawtimber 75% and Pulpwood 25% vs *Sawtimber 25% and Pulpwood 75%.*

Add Eqs. A14 and A15 to the Eq set A1 to A11, and substitute Eq. A5 for Eq. A16.

$$0.25 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = \boldsymbol{g}_{\boldsymbol{jkt}}\ \forall\, j = 1,\ k = 2,\ t = 0 \tag{A14}$$

$$0.75 \times \sum_{k}^{K} \boldsymbol{g}_{\boldsymbol{jkt}} + a_{jkt}^{+} + a_{jkt}^{-} = \boldsymbol{g}_{\boldsymbol{jkt}} \; \forall \, j = 1, \; k = 1, \; t = 0 \tag{A15}$$

$$\sum_{i}^{I} \sum_{b}^{B} \sum_{v}^{V} Z_{jitrvt} = \sum_{i}^{I} \sum_{b}^{B} \sum_{v}^{V} A_{jitrvt} \;\; \forall j \in J, \;\; t = 0 \tag{A16}$$

Appendix B. Optimal rotation

Table 1B
Land expectation value per acre for 4% and 8% discount rate.

Treatment	4% Discount Rate		8% Discount Rate	
	LEV ($/acre)	Optimal Rotation Age	LEV ($/acre)	Optimal Rotation Age
Tr.1	$710	27	$222	23
Tr.2	$645	27	$203	23
Tr.3	$729	28	$249	22
Tr.4	$760	25	$282	22

Note: LEV = Land Expectation Value at 4% and 8% discount rate. Timber prices are $7.30 per ton for pulpwood and $24.49 per ton for sawtimber. They were estimated by the inverse demand function $\boldsymbol{p}_{\boldsymbol{kt}} = \alpha_k \left(Q_{kt}^{s=d}\right)^{\beta_k}$, where α_k equals 8 for pulpwood and 30 for sawtimber, β_k is - 0.5 for pulpwood and sawtimber, and Q_{kt} is the demand.

Appendix C. Initial age distribution

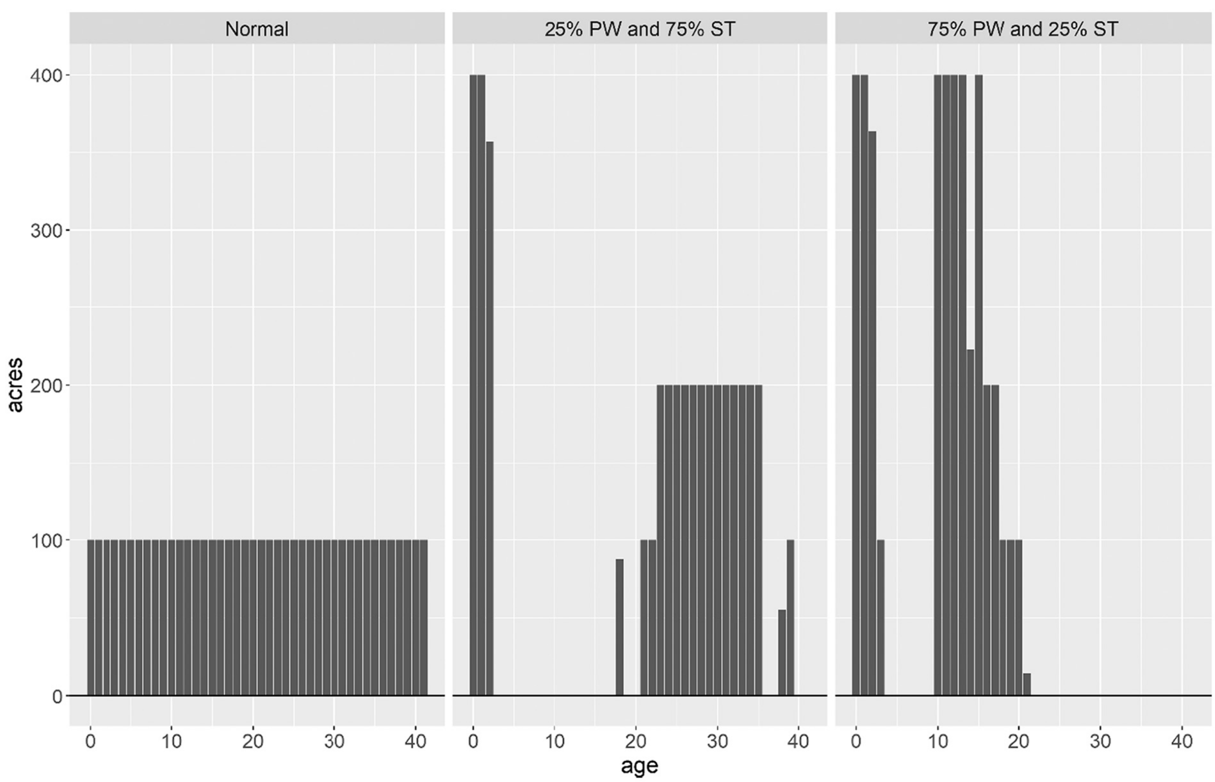

Fig. 1C. Initial age distribution by assumption. Normal age class distribution has equal area across the age 0 to 40. On the other hand, 25% PW (pulpwood) and 75% ST (sawtimber) and 75% PW and 25% ST are the results from the goal programming described in Appendix A.

Appendix D. Temporal distribution of volume (harvested and stock), and area

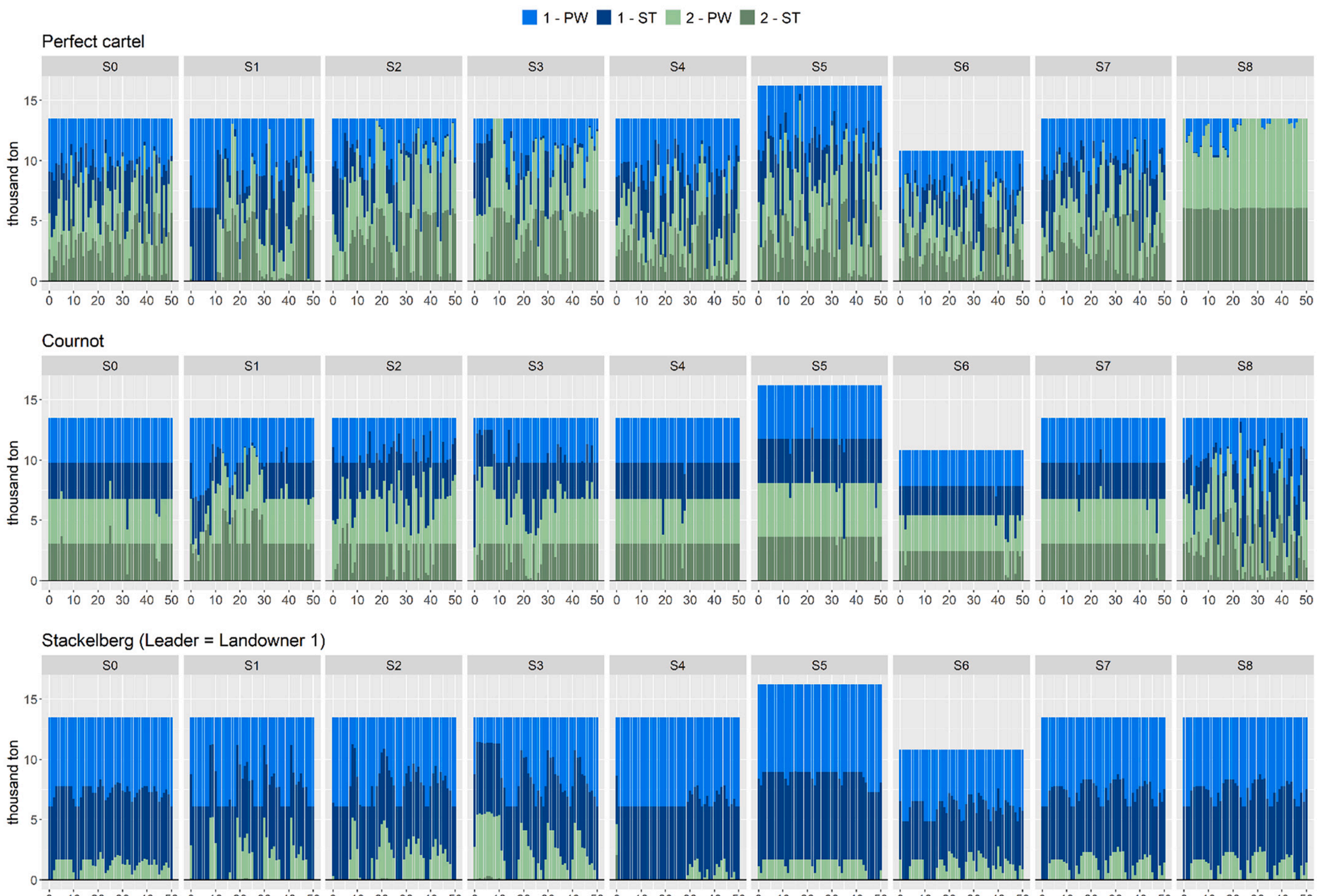

Fig. 1D. Volume harvested by products and landowners. Dark blue (green) are the volume harvested of sawtimber by landowner 1 (2), and light blue (green) is the volume harvested of pulpwood by landowner 1 (2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

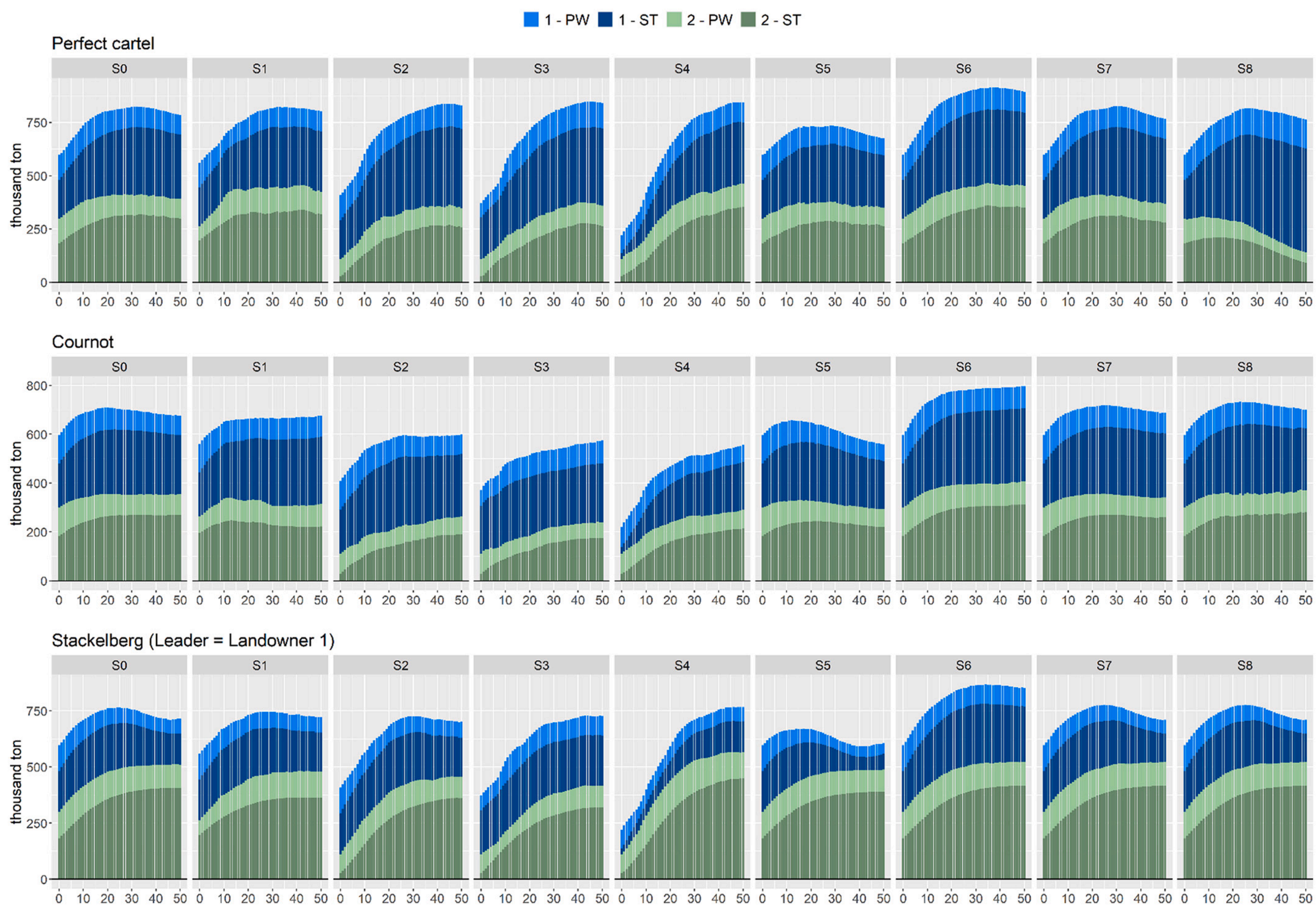

Fig. 2D. Volume Stock by products and landowners. Dark blue (green) are the stock of sawtimber in landowner 1 (2), and light blue (green) is the stock ff pulpwood in landowner 1 (2). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

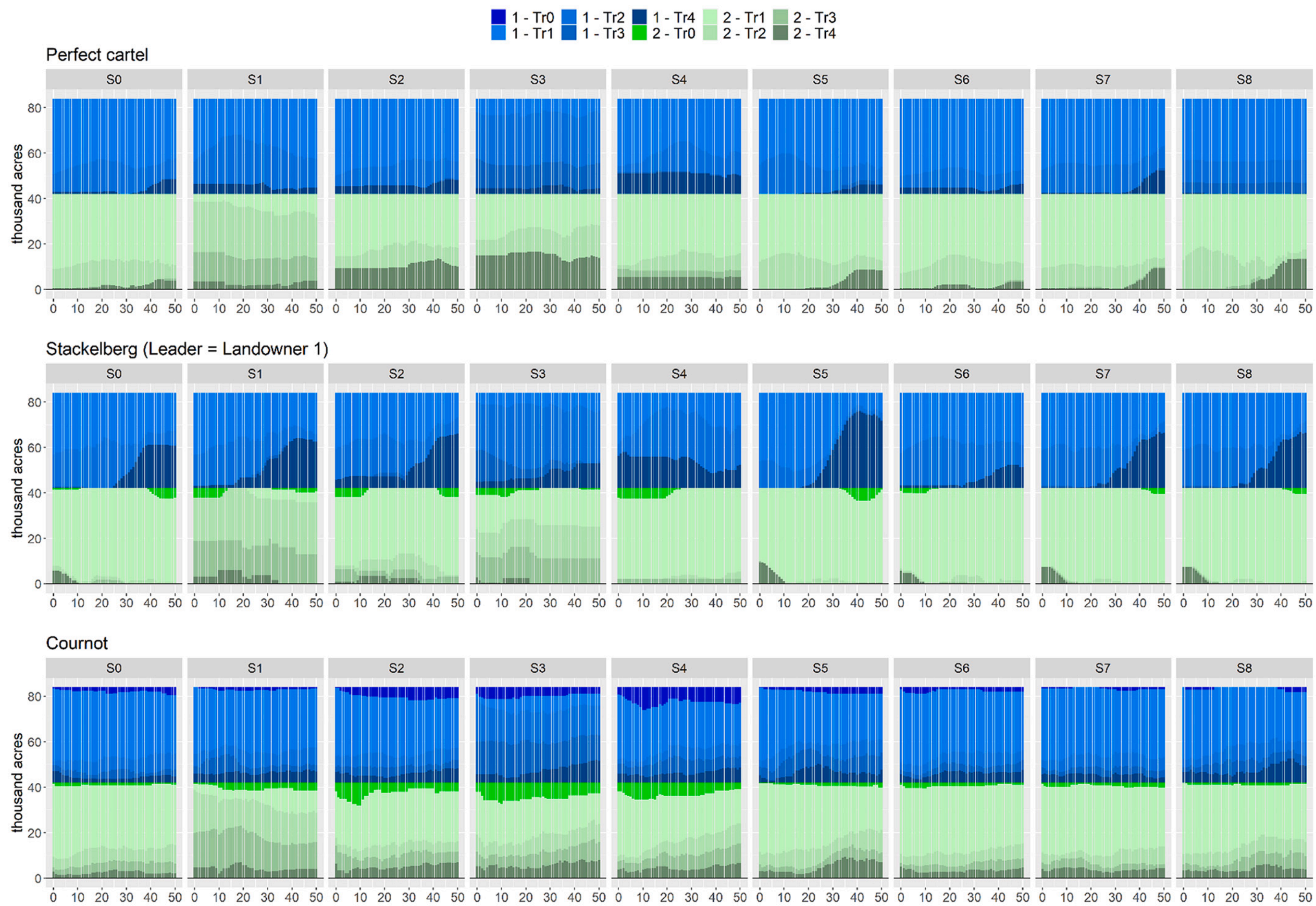

Fig. 3D. Area by treatments and landowners. Tr 0 = Not planting, Tr 1 = Clearcutting at the optimal age, Tr 2 = Thinning at 18 years old, Tr 3 = Thinning at 14 years old, Tr 4 = first thinning at 14 years old, and second thinning at 20 years old.

Appendix E. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.forpol.2022.102691.

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